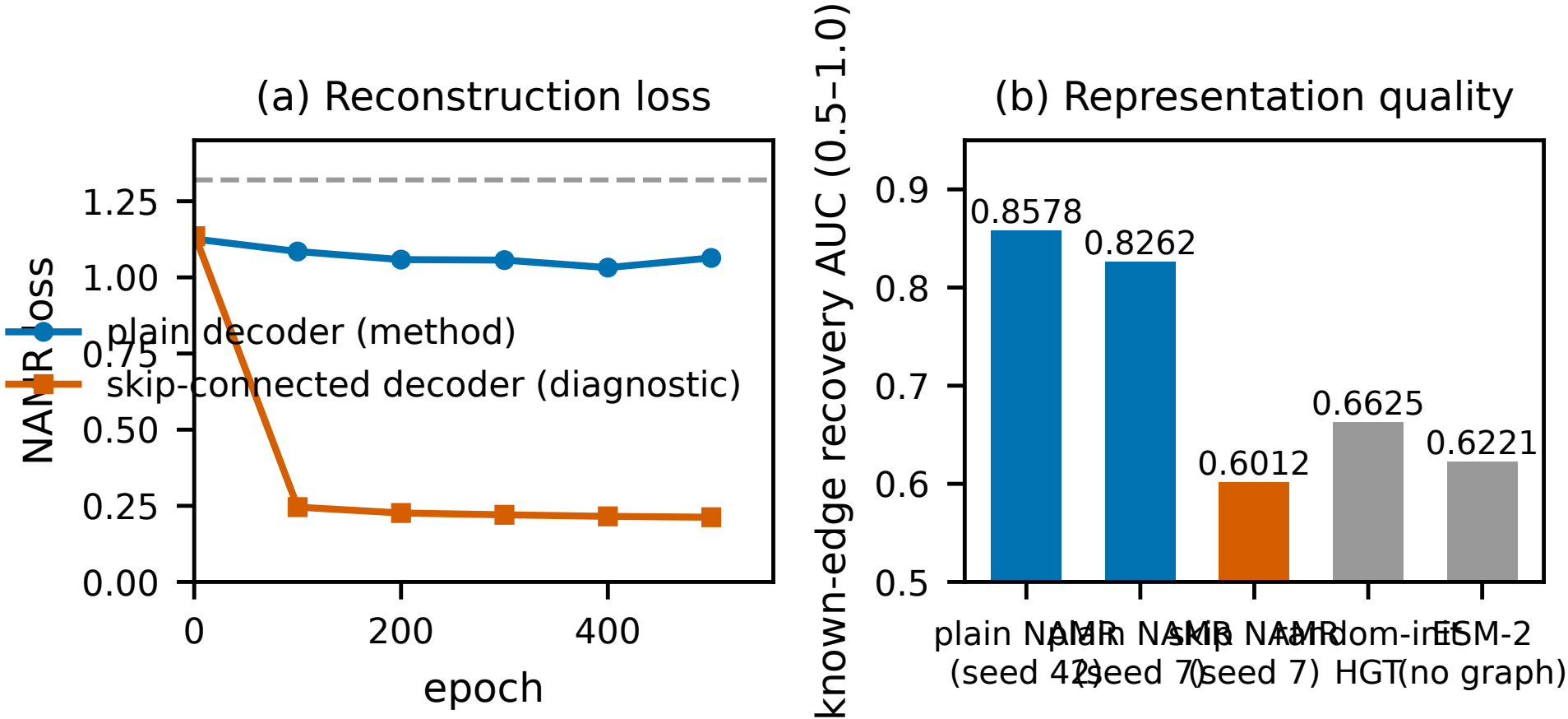
(a) Reconstruction loss
NAMR loss
epoch
plain decoder (method)
skip-connected decoder (diagnostic)
0.00
0.25
0.50
0.75
1.00
1.25
0
200
400
(b) Representation quality
known-edge recovery AUC (0.5–1.0)
0.5
0.6
0.7
0.8
0.9
0.8578
0.8262
0.6012
0.6625
0.6221
plain NAMR (seed 42)
skip NAMR (seed 7)
random-init NAMR (seed 7)
HGT
ESM-2 (no graph)